

Group 2

OUR NEW PROJECT:-

HYDROCLEAN SHIELD

- ☐ SACHIN MATHEESHA
- ☐ DINETH UPEKSHA
- ☐ PRAMOD THARAKA
- ☐ DINALI JAYAWARDHANA
- ☐ NAVARATHINAM
- RAKSHIYANANTH



HYDROCLEAN SHIELD

INTRODUCTION

NAME:-

Hydroclean Shield



CONCEPT:-

Lotus Effect



USE TECHNOLOGY:-

**Nanotechnology
Plastic Coating**



How the Technology Works

- ☐ Microscopic Papillae
- ☐ Biomimetic Coating
- ☐ Self Cleaning Mechanism

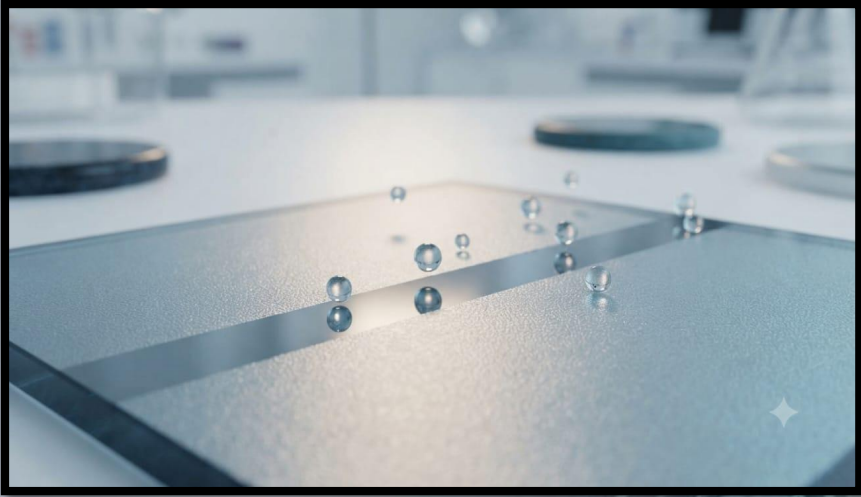


How the Material was Developed

Hydroclean Shield bridges nature and modern nanotechnology, providing revolutionary self-cleaning, anti-corrosion, and superhydrophobic solutions for a sustainable future.

BENEFITS

☐ HYDROPHOBIC



:-Water slides off with out sticking
Surface stays clean and spot free



☐ CORROSION RESISTANCE :-Prevents rust from water and air Makes the material last longer



❑ **PLASTIC COATING:-** Blocks moisture and harsh chemicals
Gives a smooth , lightweight protective layer



❑ **STRONG AND DURABLE** :-Protects against scratches and damage
Save money on repairs or replacements

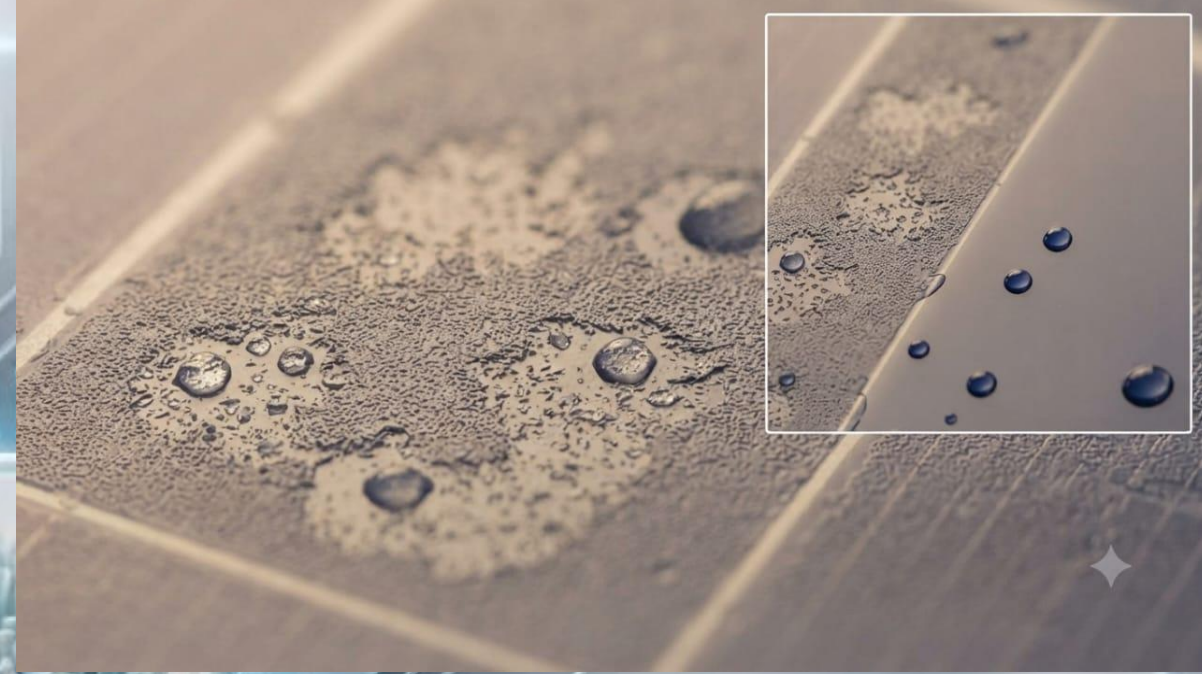


APPLICATIONS

☐ To prevent accidents

☐ To apply solar panels

☐ To apply iron materials



☐ To apply cloths

☐ To apply buildings as a paint

☐ To apply outdoor lights

Hydroclean Shield
with d'y stansy



Salt-spray Test coupons
after rust coupons



Enhanced Solar
Efficiency



Self-Cleaning
Glass



Reduced
Maintenance for
Outdoor Lights



FUTURE USE

☐ Self washing vehicles



☐ Solar panels protecting



☐ Reducing corrosion

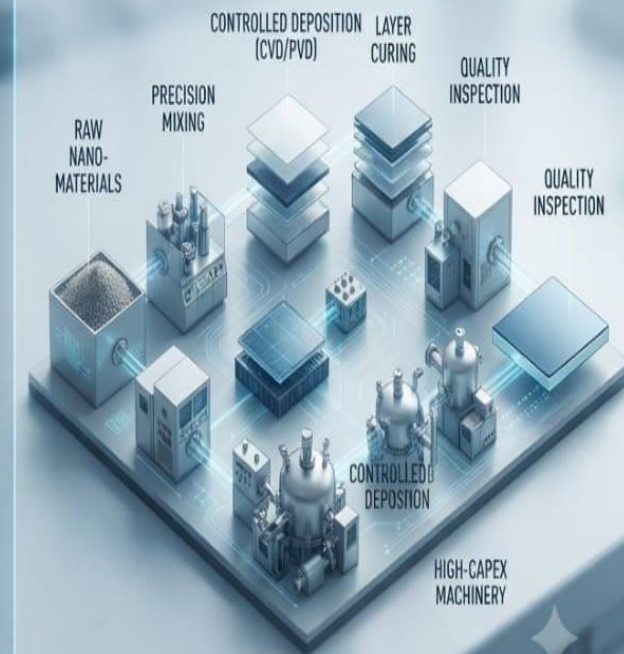
LIMITATIONS

- ❖ Durability Over Time
- ❖ Cost
- ❖ Application Complexity
- ❖ Environmental Sensitivity

CHALLENGE: DURABILITY



CHALLENGE: COST & COMPLEXITY



FUTURE RESEARCHS

❖ How to use this technology in aircrafts and ships

❖ Use to formula cars

❖ Use engines' components



Aircraft De-icing & Efficiency

Marine Anti-Fouling

HOW TO IMPROVE THAT RESEARCH ITEMS

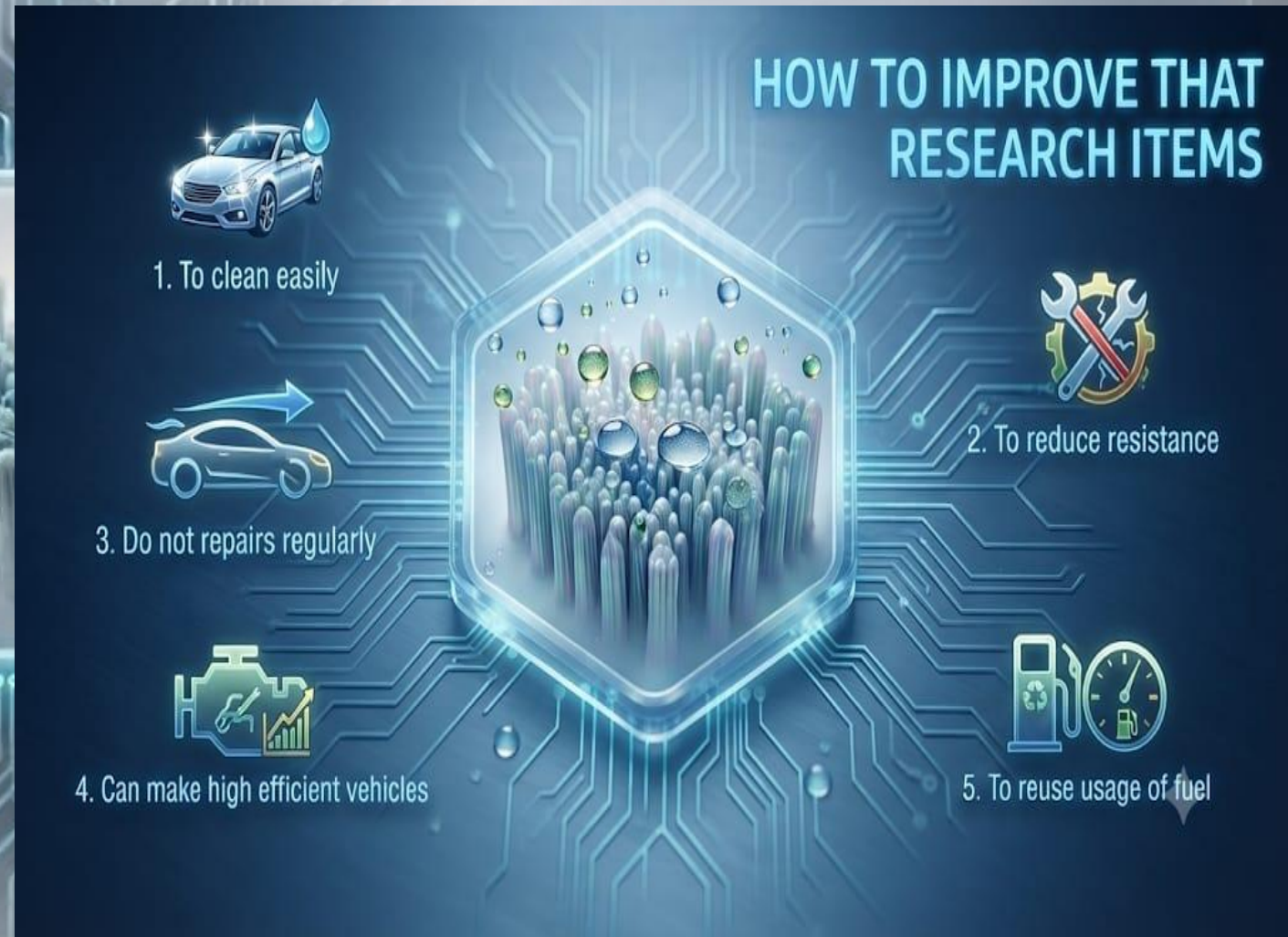
☐ To clean easily

☐ To reduce water and air resistance

☐ Do not repairs regularly

☐ Can make high efficient vehicles

☐ To reuse usage of fuel



WHY SPONSORS SHOULD INVEST IN OUR IDEA

- ❑ Very valuable technology
- ❑ Can not replace anyone
- ❑ Do not limit one country
- ❑ Eco friendly product
- ❑ Can improve various pathways



The background features a complex network of glowing blue circuit lines on a dark, textured surface. In the center, there is a cluster of stylized human figures, some of which are highlighted with glowing blue spheres. The overall aesthetic is futuristic and technological.

***THANK
YOU...!***

